

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

## Practical - VIII

|                                 |                                      |
|---------------------------------|--------------------------------------|
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| Class : SYIT                    | Batch : 1                            |
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Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

CODE:

```
# Bubble sort in Python

def bubbleSort(array):

    # loop to access each array element
    for i in range(len(array)):

        # loop to compare array elements
        for j in range(0, len(array) - i - 1):

            # compare two adjacent elements
            # change > to < to sort in descending order
            if array[j] > array[j + 1]:

                # swapping elements if elements
                # are not in the intended order
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-11, 50, 0, 30, -20]

bubbleSort(data)

print('Sorted Array in Ascending Order:')
print(data)

print("\nRudra_S033")
```

OUTPUT:

```
Sorted Array in Ascending Order:
[-20, -11, 0, 30, 50]

Rudra_S033
```

b. Insertion sort

CODE:

```
# Insertion sort in Python
def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key

data = [11, 7, 1, 5, 2]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)

print("\nRudra_So33")
```

OUTPUT:

```
Sorted Array in Ascending Order:
[1, 2, 5, 7, 11]

Rudra_So33
```

c. Selection sort

CODE:

```
# Selection sort in Python
def selectionSort(array, size):
    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):
            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)

print("\nRudra_So33")
```

OUTPUT:

Sorted Array in Ascending Order:

[-9, -2, 0, 11, 45]

Rudra\_S033

## Curiosity Questions:

### Programming Questions

1. Write a Python program to sort a list of numbers in ascending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in ascending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in ascending order using **Selection Sort**.

### Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?

7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?
10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?